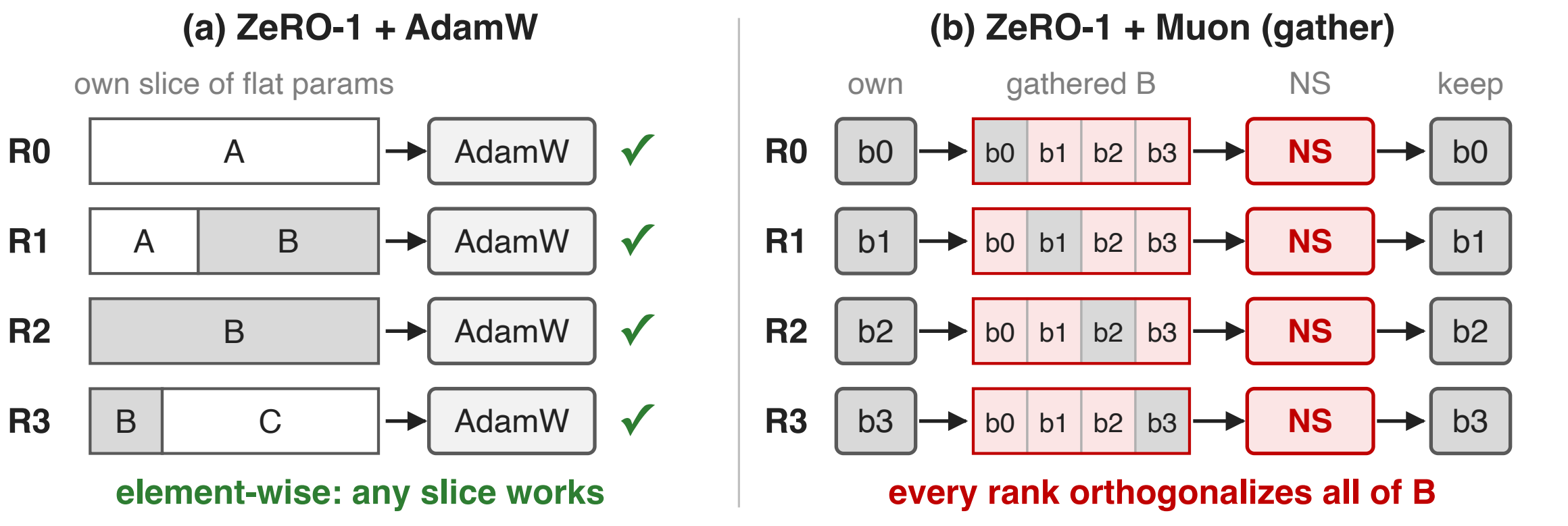
(a) ZeRO-1 + AdamW
own slice of flat params
R0
A
AdamW
R1
A
B
AdamW
R2
B
AdamW
R3
B
C
AdamW
element-wise: any slice works
(b) ZeRO-1 + Muon (gather)
own
gathered B
NS
keep
R0
b0
b0 b1 b2 b3
NS
b0
R1
b1
b0 b1 b2 b3
NS
b1
R2
b2
b0 b1 b2 b3
NS
b2
R3
b3
b0 b1 b2 b3
NS
b3
every rank orthogonalizes all of B